

LENSK, Russian Federation

WMO: 249230

Lat: 60.7250N

Lon: 114.8360E

Elev: 241

StdP: 98.46

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-46.2	-43.5	-48.5	0.0	-46.3	-45.8	0.0	-43.5	9.4	-19.7	8.5	-19.9	0.6	270	0.701

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	29.2	18.8	27.1	18.1	25.0	17.0	20.2	26.6	19.2	25.1	18.0	23.4	2.4	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	13.3	22.9	16.9	12.4	21.9	15.9	11.6	20.7	59.4	26.3	55.8	25.4	52.1	23.3	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.8	6.8	6.1	-47.1	32.9	3.3	1.6	-49.5	34.0	-51.4	34.9	-53.2	35.8	-55.6	36.9		
(5)			-47.1	22.4	3.3	1.0	-49.5	23.2	-51.4	23.8	-53.2	24.3	-55.6	25.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.1	-28.6	-24.2	-14.1	-3.1	6.2	15.4	17.7	14.1	6.1	-4.6	-19.1	-28.1
	DBStd	17.86	9.16	7.81	7.85	5.51	4.38	4.51	3.63	3.61	4.01	6.10	9.46	9.61
	HDD10.0	6076	1195	957	747	392	133	8	1	8	129	451	874	1181
	HDD18.3	8607	1453	1190	1006	642	376	109	56	136	367	709	1124	1439
	CDD10.0	573	0	0	0	0	16	169	239	137	12	0	0	0
	CDD18.3	63	0	0	0	0	0	21	36	6	0	0	0	0
	CDH23.3	818	0	0	0	0	7	309	414	87	1	0	0	0
	CDH26.7	223	0	0	0	0	1	88	119	16	0	0	0	0
Wind	WSAvg	2.6	2.9	2.7	2.6	2.7	2.9	2.4	2.1	2.1	2.6	2.8	2.9	2.7
Precipitation	PrecAvg	415	20	16	15	18	32	50	63	65	46	37	32	21
	PrecMax	557	41	29	28	42	91	111	173	135	99	76	56	46
	PrecMin	287	8	9	3	5	9	12	9	22	13	14	10	9
	PrecStd	73	9	6	6	9	19	25	39	31	25	17	12	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.7	-3.3	6.9	13.7	24.6	31.5	32.7	28.7	20.7	9.5	1.9	-3.7
		MCWB	-7.5	-4.2	1.7	5.6	13.1	18.6	20.5	19.6	13.4	5.5	0.3	-4.5
	2%	DB	-10.1	-6.8	2.5	10.8	20.6	29.1	29.8	25.8	17.7	6.6	0.0	-7.6
		MCWB	-10.8	-7.7	-1.2	4.1	10.5	17.9	19.6	17.9	11.1	3.8	-0.7	-8.2
	5%	DB	-12.4	-9.7	-0.4	8.2	18.0	27.0	27.6	23.4	14.9	4.8	-1.6	-10.6
		MCWB	-13.0	-10.4	-2.8	2.6	9.0	17.2	18.7	16.5	10.1	2.6	-2.4	-11.2
	10%	DB	-15.2	-12.4	-2.4	5.8	15.3	24.6	25.6	21.2	12.7	3.0	-4.9	-13.8
		MCWB	-15.7	-13.1	-4.5	1.1	7.9	16.0	18.1	15.5	8.9	1.3	-5.6	-14.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.5	-4.3	1.8	6.0	14.1	20.7	22.1	20.6	14.2	6.1	0.7	-4.3
		MCDB	-6.9	-3.5	6.7	11.7	21.4	27.2	29.3	26.7	18.9	7.8	1.2	-3.7
	2%	WB	-10.7	-7.8	-0.8	4.5	11.8	19.3	20.8	18.8	12.2	4.2	-0.7	-8.2
		MCDB	-10.1	-6.9	1.8	10.0	19.2	26.7	27.6	24.3	16.0	6.3	0.0	-7.6
	5%	WB	-12.9	-10.4	-2.6	3.0	9.9	18.0	19.8	17.6	10.8	2.8	-2.4	-11.1
		MCDB	-12.4	-9.7	-0.3	7.5	16.8	25.0	25.7	21.9	13.6	4.4	-1.5	-10.6
	10%	WB	-15.7	-13.0	-4.5	1.6	8.2	16.8	18.8	16.4	9.5	1.4	-5.5	-14.3
		MCDB	-15.3	-12.5	-2.6	5.2	14.8	23.1	24.1	20.2	12.6	2.8	-4.8	-13.8
Mean Daily Temperature Range	5% DB	MDBR	7.2	9.6	13.1	11.7	11.1	12.5	11.2	10.8	8.5	6.6	7.3	7.0
		MCDBR	8.7	9.1	11.9	13.8	16.7	16.7	15.6	15.7	13.0	7.7	7.6	7.9
		MCWBR	8.5	8.5	9.3	8.3	8.6	7.1	7.1	8.3	7.4	5.4	7.4	7.8
	5% WB	MCDBR	8.7	9.0	11.2	11.9	14.9	14.1	13.4	12.9	11.1	7.0	7.3	8.1
		MCWBR	8.4	8.5	8.8	7.4	8.2	6.6	6.9	7.3	6.8	5.3	7.2	7.9
Clear-Sky Solar Irradiance	taub	0.245	0.280	0.306	0.341	0.369	0.411	0.451	0.418	0.334	0.293	0.244	0.213	
	taud	1.955	2.029	2.068	2.118	2.309	2.294	2.182	2.244	2.460	2.333	2.001	1.892	
	Ebn at Noon	462	682	801	845	853	817	767	763	789	690	458	332	
	Edh at Noon	46	80	109	126	114	118	129	112	76	59	42	30	
All-Sky Solar Radiation	RadAvg	0.43	1.38	3.00	4.61	5.14	5.64	5.08	4.00	2.44	1.13	0.53	0.23	
	RadStd	0.03	0.10	0.20	0.33	0.33	0.51	0.44	0.27	0.24	0.13	0.06	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page